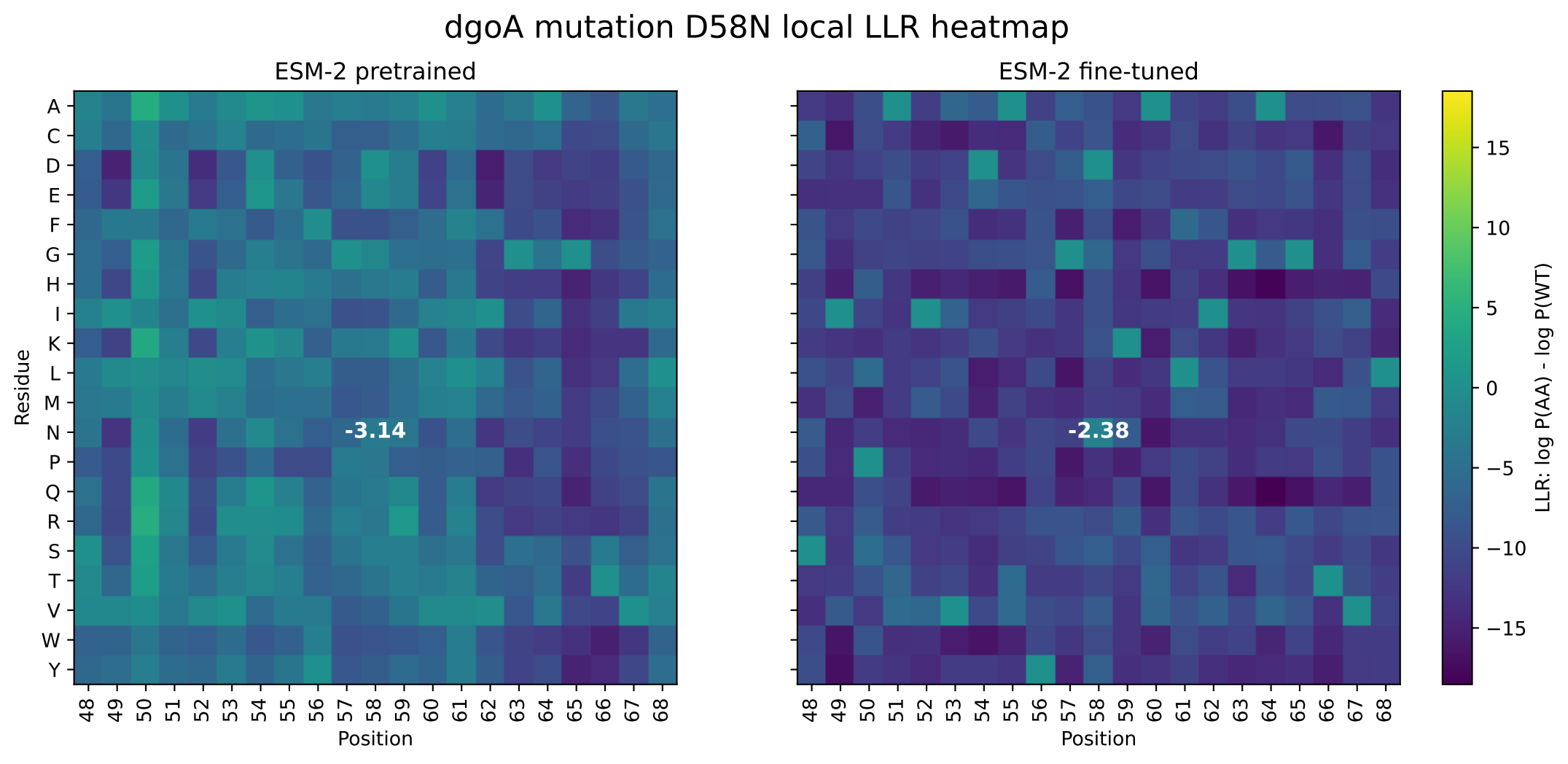
dgoA mutation D58N local LLR heatmap
ESM-2 pretrained
ESM-2 fine-tuned
-3.14
-2.38
Residue
A
C
D
E
F
G
H
I
K
L
M
N
P
Q
R
S
T
V
W
Y
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
Position
15
10
5
0
−5
−10
−15
LLR: log P(AA) - log P(WT)